Part Two

Summary of Math Properties

Arithmetic

1. A *prime number* is an integer that is divisible only by itself and 1.
2. An even number is divisible by 2, and can be written as $2x$.
3. An odd number is not divisible by 2, and can be written as $2x + 1$.
4. Division by zero is undefined.
5. Perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81 . . .
6. Perfect cubes: 1, 8, 27, 64, 125 . . .
7. If the last digit of a integer is 0, 2, 4, 6, or 8, then it is divisible by 2.
8. An integer is divisible by 3 if the sum of its digits is divisible by 3.
9. If the last digit of a integer is 0 or 5, then it is divisible by 5.
10. Miscellaneous Properties of Positive and Negative Numbers:
    A. The product (quotient) of positive numbers is positive.
    B. The product (quotient) of a positive number and a negative number is negative.
    C. The product (quotient) of an even number of negative numbers is positive.
    D. The product (quotient) of an odd number of negative numbers is negative.
    E. The sum of negative numbers is negative.
    F. A number raised to an even exponent is greater than or equal to zero.

$$\begin{aligned} &even \times even = even \\ &odd \times odd = odd \\ &even \times odd = even \end{aligned}$$

$$\begin{aligned} &even + even = even \\ &odd + odd = even \\ &even + odd = odd \end{aligned}$$

11. Consecutive integers are written as $x, x + 1, x + 2, \ldots$
12. Consecutive even or odd integers are written as $x, x + 2, x + 4, \ldots$
13. The integer zero is neither positive nor negative, but it is even: $0 = 2 \cdot 0$.
14. Commutative property: $x + y = y + x$. Example: $5 + 4 = 4 + 5$.
15. Associative property: $(x + y) + z = x + (y + z)$. Example: $(1 + 2) + 3 = 1 + (2 + 3)$.
16. Order of operations: Parentheses, Exponents, Multiplication, Division, Addition, Subtraction.
17. $-\frac{x}{y} = \frac{-x}{y} = \frac{x}{-y}$. Example: $-\frac{2}{3} = \frac{-2}{3} = \frac{2}{-3}$
18.

$33\frac{1}{3}\% = \frac{1}{3}$	$20\% = \frac{1}{5}$
$66\frac{2}{3}\% = \frac{2}{3}$	$40\% = \frac{2}{5}$
$25\% = \frac{1}{4}$	$60\% = \frac{3}{5}$
$50\% = \frac{1}{2}$	$80\% = \frac{4}{5}$

19.

$\frac{1}{100} = .01$	$\frac{1}{10} = .1$	$\frac{2}{5} = .4$
$\frac{1}{50} = .02$	$\frac{1}{5} = .2$	$\frac{1}{2} = .5$
$\frac{1}{25} = .04$	$\frac{1}{4} = .25$	$\frac{2}{3} = .666\ldots$
$\frac{1}{20} = .05$	$\frac{1}{3} = .333\ldots$	$\frac{3}{4} = .75$

20. Common measurements:
1 foot = 12 inches
1 yard = 3 feet
1 quart = 2 pints
1 gallon = 4 quarts
1 pound = 16 ounces

21. Important approximations: $\sqrt{2} \approx 1.4$ $\sqrt{3} \approx 1.7$ $\pi \approx 3.14$

22. *"The remainder is r when p is divided by q"* means $p = qz + r$; the integer z is called the quotient. For instance, *"The remainder is 1 when 7 is divided by 3"* means $7 = 3 \cdot 2 + 1$.

23. $Probability = \dfrac{number\ of\ outcomes}{total\ number\ of\ possible\ outcomes}$

Algebra

24. Multiplying or dividing both sides of an inequality by a negative number reverses the inequality. That is, if $x > y$ and $c < 0$, then $cx < cy$.

25. Transitive Property: If $x < y$ and $y < z$, then $x < z$.

26. Like Inequalities Can Be Added: If $x < y$ and $w < z$, then $x + w < y + z$.

27. Rules for exponents:

$x^a \cdot x^b = x^{a+b}$ Caution, $x^a + x^b \neq x^{a+b}$

$(x^a)^b = x^{ab}$

$(xy)^a = x^a \cdot y^a$

$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$

$\frac{x^a}{x^b} = x^{a-b}$, if $a > b$. $\frac{x^a}{x^b} = \frac{1}{x^{b-a}}$, if $b > a$.

$x^0 = 1$

28. There are only two rules for roots that you need to know for the GRE:

$\sqrt[n]{xy} = \sqrt[n]{x}\sqrt[n]{y}$ For example, $\sqrt{3x} = \sqrt{3}\sqrt{x}$.

$\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$ For example, $\sqrt[3]{\frac{x}{8}} = \frac{\sqrt[3]{x}}{\sqrt[3]{8}} = \frac{\sqrt[3]{x}}{2}$.

Caution: $\sqrt[n]{x+y} \neq \sqrt[n]{x} + \sqrt[n]{y}$.

29. Factoring formulas:

$$x(y+z) = xy + xz$$
$$x^2 - y^2 = (x+y)(x-y)$$
$$(x-y)^2 = x^2 - 2xy + y^2$$
$$(x+y)^2 = x^2 + 2xy + y^2$$
$$-(x-y) = y - x$$

30. Adding, multiplying, and dividing fractions:

$\frac{x}{y} + \frac{z}{y} = \frac{x+z}{y}$ and $\frac{x}{y} - \frac{z}{y} = \frac{x-z}{y}$ Example: $\frac{2}{4} + \frac{3}{4} = \frac{2+3}{4} = \frac{5}{4}$.

$\frac{w}{x} \cdot \frac{y}{z} = \frac{wy}{xz}$ Example: $\frac{1}{2} \cdot \frac{3}{4} = \frac{1 \cdot 3}{2 \cdot 4} = \frac{3}{8}$.

$\frac{w}{x} \div \frac{y}{z} = \frac{w}{x} \cdot \frac{z}{y}$ Example: $\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \cdot \frac{4}{3} = \frac{4}{6} = \frac{2}{3}$.

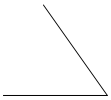
31. $x\% = \frac{x}{100}$

32. Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ are the solutions of the equation $ax^2 + bx + c = 0$.

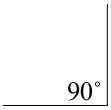
Geometry

33. There are four major types of angle measures:

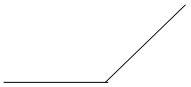
An acute angle has measure less than 90˚:



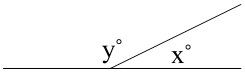
A right angle has measure 90˚:



An obtuse angle has measure greater than 90˚:

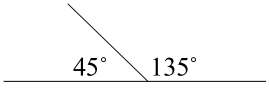


A straight angle has measure 180°:



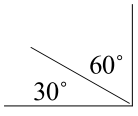
x + y = 180˚

34. Two angles are supplementary if their angle sum is 180˚:



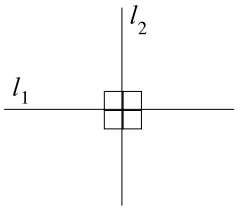
45 + 135 = 180

35. Two angles are complementary if their angle sum is 90˚:



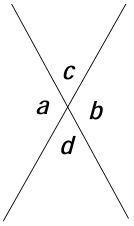
30 + 60 = 90

36. Perpendicular lines meet at right angles:



$l_1 \perp l_2$

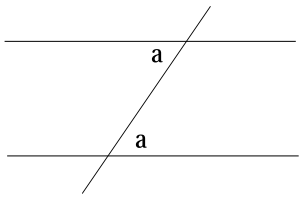
37. When two straight lines meet at a point, they form four angles. The angles opposite each other are called vertical angles, and they are congruent (equal). In the figure to the right, $a = b$, and $c = d$.



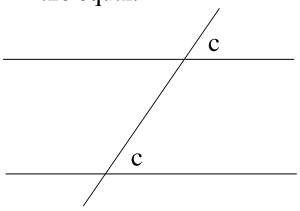
$a = b$ and $c = d$

38. When parallel lines are cut by a transversal, three important angle relationships exist:

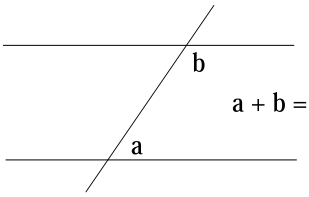
Alternate interior angles are equal.



Corresponding angles are equal.

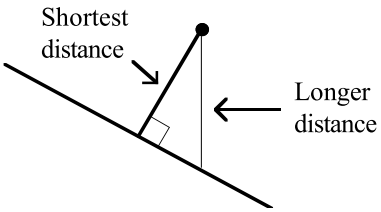


Interior angles on the same side of the transversal are supplementary.

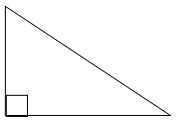


$a + b = 180^\circ$

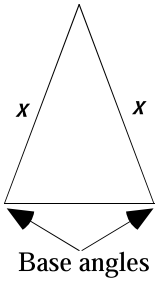
39. The shortest distance from a point not on a line to the line is along a perpendicular line.



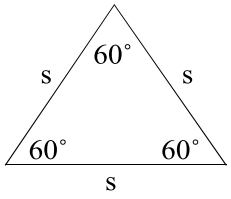
40. A triangle containing a right angle is called a *right triangle*. The right angle is denoted by a small square:



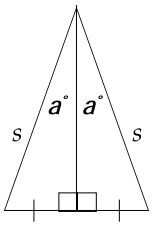
41. A triangle with two equal sides is called isosceles. The angles opposite the equal sides are called the base angles:

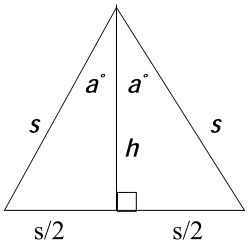


42. In an equilateral triangle, all three sides are equal and each angle is 60°:



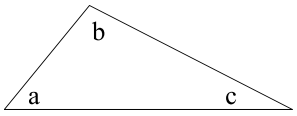
43. The altitude to the base of an isosceles or equilateral triangle bisects the base and bisects the vertex angle:

Isosceles: 

Equilateral: 

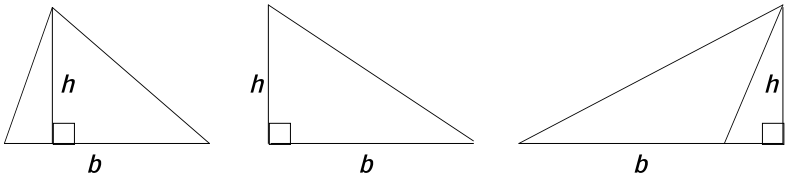
$h = \frac{s\sqrt{3}}{2}$

44. The angle sum of a triangle is 180°:



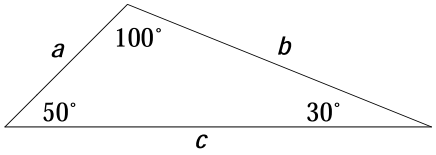
$a + b + c = 180^\circ$

45. The area of a triangle is $\frac{1}{2}bh$, where b is the base and h is the height.



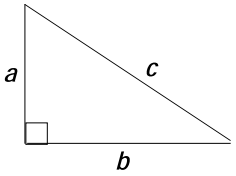
$A = \frac{1}{2}bh$

46. In a triangle, the longer side is opposite the larger angle, and vice versa:



50° is larger than 30°, so side b is longer than side a.

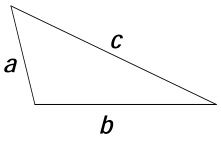
47. Pythagorean Theorem (right triangles only): The square of the hypotenuse is equal to the sum of the squares of the legs.

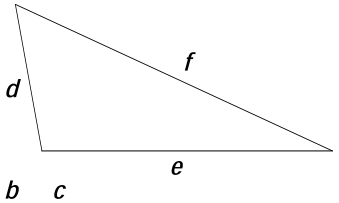


$c^2 = a^2 + b^2$

48. A Pythagorean triple: the numbers 3, 4, and 5 can always represent the sides of a right triangle and they appear very often: $5^2 = 3^2 + 4^2$.

49. Two triangles are similar (same shape and usually different size) if their corresponding angles are equal. If two triangles are similar, their corresponding sides are proportional:

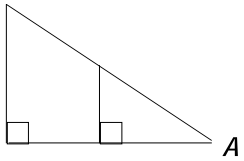




$$\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$$

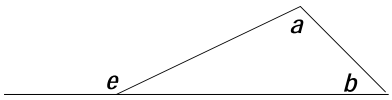
50. If two angles of a triangle are congruent to two angles of another triangle, the triangles are similar.

In the figure to the right, the large and small triangles are similar because both contain a right angle and they share $\angle A$.



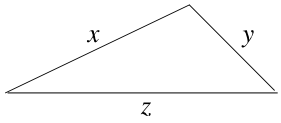
51. Two triangles are congruent (identical) if they have the same size and shape.

52. In a triangle, an exterior angle is equal to the sum of its remote interior angles and is therefore greater than either of them:



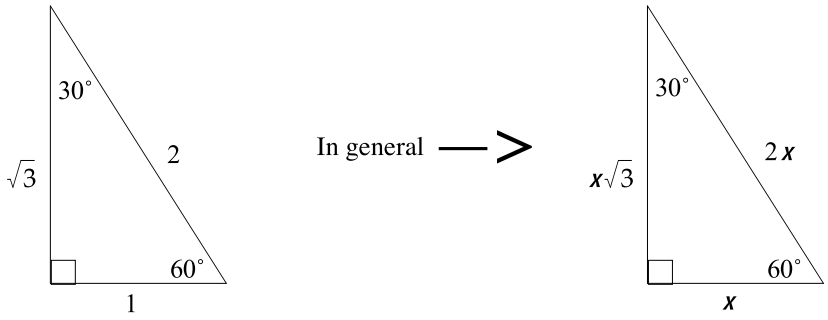
$e = a + b$ and $e > a$ and $e > b$

53. In a triangle, the sum of the lengths of any two sides is greater than the length of the remaining side:

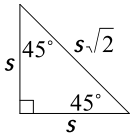


$x + y > z$
$y + z > x$
$x + z > y$

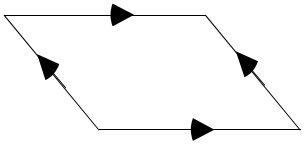
54. In a 30°–60°–90° triangle, the sides have the following relationships:



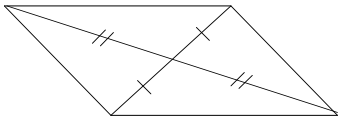
55. In a 45°–45°–90° triangle, the sides have the following relationships:



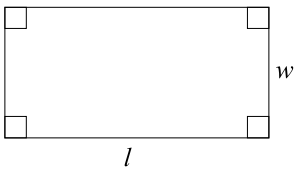
56. Opposite sides of a parallelogram are both parallel and congruent:



57. The diagonals of a parallelogram bisect each other:

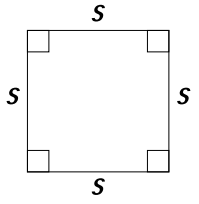


58. A parallelogram with four right angles is a *rectangle*. If w is the width and l is the length of a rectangle, then its area is $A = lw$ and its perimeter is $P = 2w + 2l$:



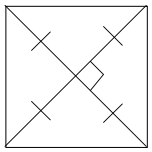
$A = l \cdot w$
$P = 2w + 2l$

59. If the opposite sides of a rectangle are equal, it is a square and its area is $A = s^2$ and its perimeter is $P = 4s$, where s is the length of a side:

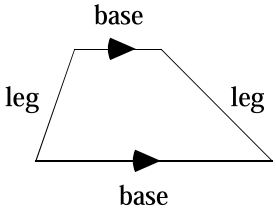


$A = s^2$
$P = 4s$

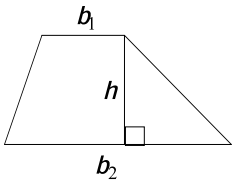
60. The diagonals of a square bisect each other and are perpendicular to each other:



61. A quadrilateral with only one pair of parallel sides is a *trapezoid*. The parallel sides are called *bases*, and the non-parallel sides are called *legs*:

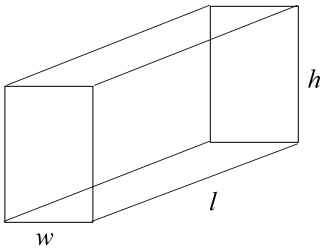


62. The area of a trapezoid is the average of the bases times the height:



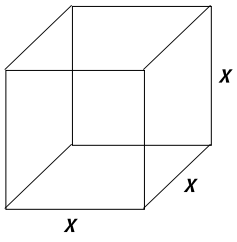
$$A = \left(\frac{b_1 + b_2}{2}\right)h$$

63. The volume of a rectangular solid (a box) is the product of the length, width, and height. The surface area is the sum of the area of the six faces:



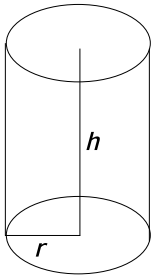
$$V = l \cdot w \cdot h$$
$$S = 2wl + 2hl + 2wh$$

64. If the length, width, and height of a rectangular solid (a box) are the same, it is a cube. Its volume is the cube of one of its sides, and its surface area is the sum of the areas of the six faces:



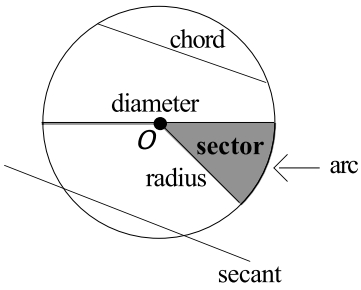
$$V = x^3$$
$$S = 6x^2$$

65. The volume of a cylinder is $V = \pi r^2 h$, and the lateral surface (excluding the top and bottom) is $S = 2\pi rh$, where r is the radius and h is the height:

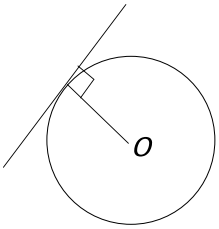


$$V = \pi r^2 h$$
$$S = 2\pi rh + 2\pi r^2$$

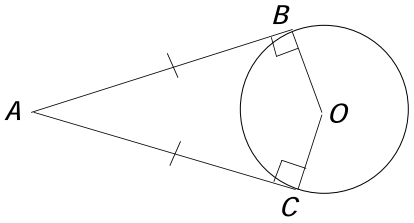
66. A line segment form the circle to its center is a *radius*.
A line segment with both end points on a circle is a *chord*.
A chord passing though the center of a circle is a *diameter*.
A diameter can be viewed as two radii, and hence a diameter's length is twice that of a radius.
A line passing through two points on a circle is a *secant*.
A piece of the circumference is an *arc*.
The area bounded by the circumference and an angle with vertex at the center of the circle is a *sector*.



67. A tangent line to a circle intersects the circle at only one point. The radius of the circle is perpendicular to the tangent line at the point of tangency:

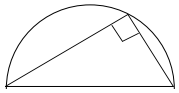


68. Two tangents to a circle from a common exterior point of the circle are congruent:

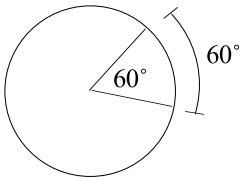


$AB \cong AC$

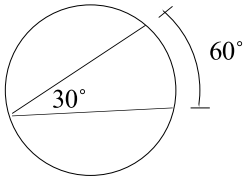
69. An angle inscribed in a semicircle is a right angle:



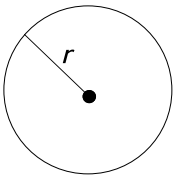
70. A central angle has by definition the same measure as its intercepted arc.



71. An inscribed angle has one-half the measure of its intercepted arc.



72. The area of a circle is πr^2, and its circumference (perimeter) is $2\pi r$, where r is the radius:



$A = \pi r^2$
$C = 2\pi r$

73. To find the area of the shaded region of a figure, subtract the area of the unshaded region from the area of the entire figure.

74. When drawing geometric figures, don't forget extreme cases.

Miscellaneous

75. To compare two fractions, cross-multiply. The larger product will be on the same side as the larger fraction.

76. Taking the square root of a fraction between 0 and 1 makes it larger.

    Caution: This is not true for fractions greater than 1. For example, $\sqrt{\frac{9}{4}} = \frac{3}{2}$. But $\frac{3}{2} < \frac{9}{4}$.

77. Squaring a fraction between 0 and 1 makes it smaller.

78. $ax^2 \neq (ax)^2$. In fact, $a^2x^2 = (ax)^2$.

79. $\frac{1/a}{b} \neq \frac{1}{a/b}$. In fact, $\frac{1/a}{b} = \frac{1}{ab}$ and $\frac{1}{a/b} = \frac{b}{a}$.

80. $-(a + b) \neq -a + b$. In fact, $-(a + b) = -a - b$.

81. $\textit{percentage increase} = \frac{\textit{increase}}{\textit{original amount}}$

82. Systems of simultaneous equations can most often be solved by merely adding or subtracting the equations.

83. When counting elements that are in overlapping sets, the total number will equal the number in one group plus the number in the other group minus the number common to both groups.

84. The number of integers between two integers inclusive is one more than their difference.

85. Elimination strategies:
    A. On hard problems, if you are asked to find the least (or greatest) number, then eliminate the least (or greatest) answer-choice.
    B. On hard problems, eliminate the answer-choice "not enough information."
    C. On hard problems, eliminate answer-choices that merely repeat numbers from the problem.
    D. On hard problems, eliminate answer-choices that can be derived from elementary operations.
    E. After you have eliminated as many answer-choices as you can, choose from the more complicated or more unusual answer-choices remaining.

86. To solve a fractional equation, multiply both sides by the LCD (lowest common denominator) to clear fractions.

87. You can cancel only over multiplication, not over addition or subtraction. For example, the c's in the expression $\frac{c + x}{c}$ cannot be canceled.

88. The average of N numbers is their sum divided by N, that is, $\textit{average} = \frac{\textit{sum}}{N}$.

89. *Weighted average:* The average between two sets of numbers is closer to the set with more numbers.

90. $\textit{Average Speed} = \frac{\textit{Total Distance}}{\textit{Total Time}}$

91. $\textit{Distance} = \textit{Rate} \times \textit{Time}$

92. *Work* = *Rate* × *Time*, or $W = R \times T$. The amount of work done is usually 1 unit. Hence, the formula becomes $1 = R \times T$. Solving this for R gives $R = \frac{1}{T}$.

93. *Interest* = *Amount* × *Time* × *Rate*

94. Principles for solving quantitative comparisons
    A. You can add or subtract the same term (number) from both sides of a quantitative comparison problem.
    B. You can multiply or divide both sides of a quantitative comparison problem by the same positive term (number). (Caution: this cannot be done if the term can ever be negative or zero.)
    C. When using substitution on quantitative comparison problems, you must plug in all five major types of numbers: positives, negatives, fractions, 0, and 1. Test 0, 1, 2, –2, and 1/2, in that order.
    D. If there are only numbers (i.e., no variables) in a quantitative comparison problem, then "not-enough-information" cannot be the answer.

95. Substitution (Special Cases):
    A. In a problem with two variables, say, x and y, you must check the case in which $x = y$. (This often gives a double case.)
    B. When you are given that $x < 0$, you must plug in negative whole numbers, negative fractions, and –1. (Choose the numbers –1, –2, and –1/2, in that order.)
    C. Sometimes you have to plug in the first three numbers (but never more than three) from a class of numbers.